

C Operators

Operator	Name	Example	Defined
Arithmetic Operators			
*	Multiplication	x*y	Multiply x times y
/	Division	x/y	Divide x by y
%	Modulo	x%y	Provide the remainder of x divided by y
+	Addition	x+y	Add x and y
-	Subtraction	x-y	Subtract y from x
++	Increment	x++	Increment x after using it
--	Decrement	--x	Decrement x before using it
-	Negation	-x	Multiply x by -1
+	Unary Plus	+x	Show x is positive
Relational and Logical Operators			
>	Greater than	x>y	1 if x is greater than y, otherwise 0
>=	Greater than or equal to	x>=y	1 if x is greater than or equal to y, otherwise 0
<	Less than	x<y	1 if x is less than y, otherwise 0
<=	Less than or equal to	x<=y	1 if x is less than or equal to y, otherwise 0
==	Equal to	x==y	1 if x equals y, otherwise 0
!=	Not equal to	x!=y	1 if x is not equal to y, otherwise 0
!	Logical NOT	!x	1 if x is 0, otherwise 1
&&	Logical AND	X&&Y	0 if either x or y is, otherwise 1
	Logical OR	x y	0 if both x and y are 0, otherwise 1
Bitwise Operators			
~	Bitwise Complement	~x	Changes 1 bits to and 0 bits 1
&	Bitwise AND	X&y	Bitwise and of x and y
	Bitwise OR	x y	Bitwise OR of x and y
^	Bitwise XOR	x^y	Bitwise XOR of x and y
<<	Left Shift	x<<2	Bits in x are shifted left 2 bit positions
>>	Right Shift	x>>3	Bits in x are shifted right 3 bit positions
Data Access and Size Operators			
[]	Array Element	X[6]	Seventh element of array x
.	Member Selection	PORTD.2	Bit 2 of Port D
->	Member Selection	pStruct->x	Member x of the structure pointed to by pStruct
*	Indirection	*p	Contents of memory located at address p
&	Address of	&x	Address of the variable x
Assignment Operators			
=	Assignment	x=y	The value in y is to variable x
+=	Compound Assignment	x+=x+y	x is assigned the sum of x and y
-=	Compound Assignment	x-=x-y	x is assigned the difference between x and y
=	Compound Assignment	x=x*y	x is assigned the value of x multiplied by y
/=	Compound Assignment	x/=x/y	x is assigned the value of x divided by y
%=	Compound Assignment	x%=x%y	x is assigned the value of the modulus of x and y
&=	Compound Assignment	x=x&y	x is assigned the value of x AND with y
^=	Compound Assignment	x=x^y	x is assigned the value of x XOR with y
=	Compound Assignment	x=x y	x is assigned the value of x OR with y
<<=	Compound Assignment	x=x<<y	x is assigned the result of left shifting x by y bits
>>=	Compound Assignment	x=x>>y	x is assigned the result of right shifting x by y bits
Miscellaneous Operators			
()	Function	Wait(10)	Call wait function with argument 10
(type)	Type cast	(double)x	x converted to a double
?:	Conditional	X?y:z	If x is not 0 evaluate y, otherwise evaluate z
,	Sequential evaluation	x++,y++	Increment x first, then increment y

Operator Precedence	
Precedence	Operator
1	() [] . ->
2	++ -- ! ~ + - * & sizeof
3	Cast operator
4	* / %
5	+ -
6	<< >>
7	< <= > >=
8	== !=
9	&
10	^
11	
12	&&
13	
14	? : (conditional operator)
15	= += -= *= /= %= &= ^= = <<= >>=
16	, (comma operator)

When more than one operator is used in a C statement, they will be evaluated in the order that they appear in this table from top to bottom.